6	**(a)**		Radius = 4 $(x-10)^2+(y-4)^2=16$	**B1** **B1** **[2]**	**3.1a** **1.1**	soi by 16 o.e.	
6	**(b)**		Where $y=x$ meets the circle $(x-10)^2+(x-4)^2=16$ $2x^2-28x+116=16$ or $x^2-14x+50=0$ or $2x^2-28x+100=0$ $b^2-4ac=-4$ or -16 so no meeting points i.e. not a tangent	**M1** **M1** **M1** **E1**	**3.1a** **1.1** **1.1** **2.4**	For sub'n of $y=x$ into their circle equation For expanding and collecting like terms Rearranging to 3 term quadratic =0 Or $(x-7)^2+1=0$ no meeting points hence not a tangent	'No real roots' not enough for E1 Allow E1 for correct reasoning following omission of '=0'
			Alternative method Angle between $y=x$ and x-axis = 45º Let θ be the angle between the x-axis and the line joining (0, 0) and (10, 4). tan θ =0.4 θ= 21.80(2 d.p.) If $y=x$ was a tangent θ would be 22.5º hence not a tangent	**M1** **M1** **M1** **E1**		**Alternative method 2** gradient of normal is -1 Line joining point of contact (k, k) to centre (10, 4) must have gradient -1 $k=7$ (7, 7) does not lie on circle	
				[4]			
6	**(c)**		$5\mathbf{i}+2\mathbf{j}$ or $\begin{pmatrix}5\\2\end{pmatrix}$	**B1** **[1]**	**2.5**	Must be correct vector notation; not (5, 2)	